PulsePoint Data Pipeline – Ingestion Layer (README)

1. Project Overview

This document outlines the data ingestion layer of the PulsePoint pipeline. The goal of this stage is to integrate multiple wearable sensor datasets, standardize and clean the data, ingest the data into AWS using a serverless architecture, and store the data in a centralized raw data lake (S3). This stage serves as the foundation for downstream machine learning, processing, and dashboard development.

2. Pipeline Architecture

Colab (Data Cleaning & Integration) → API Gateway (POST /ingest endpoint) → AWS Lambda (processing & ingestion) → Amazon S3 (raw data storage)

3. What Has Been Implemented

Data Integration: Combined UCI Heart, PMData, and WESAD datasets into a unified schema.

Data Cleaning: Handled missing values and converted NaN to null for JSON compatibility.

Payload Generation: Exported data as JSON batches with batch_id and records.

AWS Ingestion Pipeline: Configured API Gateway, Lambda, and S3 raw zone storage.

4. AWS Resources

API Gateway Endpoint:

`https://0tcvknirlc.execute-api.us-east-1.amazonaws.com/ingest`

Lambda Function: pulsepoint-ingestion-lambda

S3 Bucket: pulsepoint-raw-zone-mena (raw/)

5. Data Format Specification

```
{
```

```
  "batch_id": "pulsepoint_batch_001",
  "records": [
    {
      "record_id": 1,
      "participant_id": "H1",
      "source_dataset": "uci_heart",
      "age": 63,
      "sex": 1,
      "resting_bp": 145,
      "chol": 233,
      "max_hr": 150,
      "avg_hr": null,
      "avg_steps": null,
      "avg_calories": null,
      "avg_sleep_score": null,
      "eda_mean": null,
      "bvp_mean": null,
      "temp_mean": null,
      "ecg_mean": null,
      "cardio_risk_label": 1,
      "stress_label_mean": null
    }
  ]
}
```


6. How to Access the Data

Use sample_data.json for real pipeline output.

Use payload_format.json as schema reference.

7. Shared Files

sample_data.json, payload_format.json, lambda_function.py, README.docx

8. Access Considerations

Due to AWS Learner Lab limitations, direct IAM sharing may not be possible. Use shared files and documentation for collaboration.

9. Key Design Decisions

Serverless architecture, JSON format, batch ingestion, separation of raw and processed layers.

10. Summary

The ingestion layer is fully implemented and operational. The pipeline is ready for PCA, logistic regression, AWS Fargate deployment, and Streamlit dashboard development.